

DCDH Real Estate Platform

PHASE 0 - MVP

DCDH Real Estate Platform: Phase 0 - MVP Summary

This document outlines the scope, technology stack and requirements for the Minimum Viable Product (MVP) of the DCDH Real Estate Platform.

Technology Stack

- **Frontend:** Next.js
- **Backend:** Node.js with Express.js (for RESTful API)
- **Database:** PostgreSQL
- **Authentication:** Email + OTP based
- **Cloud & Storage:** AWS/DigitalOcean for hosting, with S3 for media storage.
- **Maps:** Google Maps API (for location and property mapping)

Project Scope: Phase 1 (MVP)

The MVP is structured around seven core components:

1. Business Brand Website

- **Purpose:** Professional, public-facing entry point for all users.
- **Key Features:**
 - Hero section with search bar and value proposition.
 - Featured properties showcase on the homepage.
 - Informational pages: About Us, How It Works, Contact Us, Terms & Conditions, Privacy Policy.
 - Property category navigation (Buy, Rent, Commercial).
 - Responsive design.
 - Property details pages.

2. Authentication System

- **Purpose:** Secure user login and access control.
- **Key Features:**
 - Email + OTP (One-Time Password) based authentication.
 - Role-based access control.

3. Admin Portal

- **Purpose:** Centralized dashboard for platform administration.
- **Key Features:**
 - **Leads Management:** View, manage, lead dashboards (charts, graphs), and

Kanban board for inquiries.

- **Properties Management:** Add, review, approve, reject, edit, or delete property listings.
- **Users Management:** View, manage, suspend, or activate user accounts.
- Dashboard with key metrics and analytics.
- Property and user verification/approval workflows.

4. Owner Portal

- **Purpose:** Dedicated area for property owners to manage their listings.
- **Key Features:**
 - **Owner Onboarding:** Step-by-step registration, profile setup, and document upload for verification.
 - **Property Posting:** Add new listings with detailed information (type, area, price, amenities), multiple media uploads (images, videos), and location for Google Maps display.
 - View all listings with status (Active, Pending Approval, Rejected).
 - Owner dashboard with listing overview.

5. Search Functionality

- **Purpose:** Search capabilities for property discovery.
- **Key Features:**
 - Filtering by property type (Residential, Commercial, Land).
 - Filtering by transaction type (Buy, Rent).
 - Price range filtering.
 - Category-based browsing (Apartments, Villas, Offices, etc.).
 - Sorting by price and date posted.
 - Pagination for search results.

6. Dynamic Property Pages

- **Purpose:** Individual, comprehensive detail pages for each listing.
- **Key Features:**
 - SEO-friendly URLs.
 - High-quality image gallery.
 - Complete property details, specifications, amenities, and description.
 - Google Maps integration showing property location and nearby landmarks.
 - Inquiry form for interested parties (generates Leads).
 - Social sharing buttons.

7. Infrastructure & Deployment

- **Purpose:** Complete cloud setup with automated deployment processes.
- **Key Components:**
 - AWS/DigitalOcean server setup and configuration.
 - PostgreSQL database setup with backups.
 - S3/Object Storage for media files.
 - Domain configuration and SSL certificate setup.
 - Environment configuration (staging and production).

Here is a simplified view of the DCDH Real Estate Platform MVP architecture using the terms you mentioned:

Component	Acronym	Technology Stack	Key Responsibilities
Frontend	FE	Next.js	User interface (Business Website, Admin Portal, Owner Portal), routing, responsive design, consumption of API.
Backend	BE	Node.js with Express.js	Core business logic, RESTful API development, data validation, authentication processing, and communication with the database and storage.
Database	DB	PostgreSQL	Storage for structured data (user accounts, property details, leads).
Storage	(Part of DB)	AWS S3 / DigitalOcean Object Storage	Storage for unstructured media files (property images and videos).
Authentication	Auth	Email + OTP based	Secure user login and role-based access control.
Mapping	Map	Google Maps API	Location display, property mapping, and landmark integration.

8. Development Assets Requirements

- **DCDH Development Gmail ID:** A new Gmail ID will be created to serve as the central administrative email for all development-related services, accounts, and documentation.
 - **A Claude subscription**
 - **Access to the domain, DNS and any servers available.**
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Proposed Changes :

→ **Authentication:** WhatsApp OTP based - **MSG91**

→ **Tenant Portal (Lite) :** Tenant Dashboard, Tenant Referral Dashboard, Tenant Wallet, able to view properties and raise enquiries. **(Login is Mandatory for Raising Enquiries).**

→ **Referral Program : “Refer a Friend” / “Refer a Property”**

Refer a Friend (Tenant) : Any user should be able to Refer a friend (Tenant) using the website. The user will just **write the WhatsApp number** and click on **“Refer a Friend”**. Then Login Link will be sent to the Referred user on WhatsApp by WhatsApp API Tool.

Refer a Property : It will have options like **Property Category, Property type, Property Photos/videos/To-Let Board (optional), Owner Contact Number (Mandatory), Property Location (optional).**

→ **Upcoming Features Section :** For each feature, requests will be sent to WhatsApp API Automation according to the feature requested !! Then, WhatsApp API Automation will handle the rest !

Architect

Interior Designers

Book a P.G. (Hourly, Weeks, Month)

Book Farm House / Resorts (Hourly, Day, Weeks)

Book Co-working space (Hourly, Daily, Weekly, Monthly)